

Many workers. One writer. A durable memory.

SQLite on one machine / claiming work and saving results are separate operations

A CLAIM THE WORK



Conditional UPDATE: only a pending item can be claimed.
Heartbeat renews a 3 h lease every 5 min; stale work returns to pending.

B SERIALIZE THE WRITES



Busy writer? Wait up to 30 s; retry with backoff and jitter (10 attempts total).

C KEEP TWO KINDS OF MEMORY

batch_jobs → batch_items
queue state · owner · lease · progress

analysis_runs → analysis_results
run config · year · validated answer

Per-year checkpoints reduce repeated work. A unique index rejects duplicate rows.
SQLite backup API keeps five copies. Leases are not an exactly-once guarantee.